Restaurant Spending Analysis

Plotly Express tips dataset | Python | Customer Analytics

Objective

Examine how restaurant spending varies by day, smoking status and meal period, and assess whether party size is associated with total bill.

Dataset and methods

The dataset contains 244 restaurant bills with total_bill, tip, sex, smoker, day, time and size. Methods include descriptive statistics, Welch independent-samples t-test, Chi-square testing, simple linear regression and residual diagnostics.

Analysis workflow

The project combines descriptive group comparisons with formal hypothesis tests. A categorical bill-size variable is used for the Chi-square analysis, while simple linear regression evaluates the relationship between party size and total bill.

Business interpretation

Party size is relevant to table allocation, expected spend and capacity planning. Group differences should be interpreted as associations, not causal effects.

Limitations

The dataset is a compact observational sample, so results should not be treated as causal or automatically generalised to all restaurants. Regression diagnostics and robust standard errors are included as sensitivity checks.

Reproducibility

The project is designed as a compact Python analytics portfolio example covering cleaning, descriptive analysis, hypothesis testing, regression and model diagnostics. The repository includes the analysis script, requirements and supporting figures.